Second Midterm Exam - Solutions

Name: _____

Please circle your section number!

Section 1	Section 2	Section 3	Section 4
Chenxi Wu (TR 1:00 PM)	Niclas Technau (MWF 1:20 PM)	Ander Aguirre (TR 9:30 AM)	Sarah Tammen (MWF 12:05 PM)

- **There are 6 problems on the exam, some of them have multiple parts.**
- Read the problems carefully and budget your time wisely.
- If you are running out of time, please at least say *how* you would solve a given problem, preferably using notation, even if you do not have the time to do so. This may earn you partial credit.
- No calculators or other electronic devices, please. **Turn off your phone.**
- You do not have to carry out complicated numerical computations, but you should simplify your answer if it is possible with reasonable effort. In particular, geometric series must be evaluated.
- Please present your solutions in a clear manner. **Justify your steps.** Numerical answers without explanation cannot get credit. Cross out the writing that you do not wish to be graded on.
- You can use the empty pages for additional space for writing if necessary.

Question	Points	Score
1	18	
2	10	
3	20	
4	19	
5	15	
6	18	
Total:	100	

1. In any parts of the following problem for which you deem it relevant, you may leave answers in terms of the function Φ (the CDF of a standard normal random variable) or its inverse.

   (a) (5 points) The dean of a certain university wants to estimate the proportion of students that are left-handed. He plans to send a survey to a random sample of the student body. If the dean wants to obtain a 95% confidence interval for the proportion of students that are left-handed and he wants an interval of width ε, then how many students should he survey? Give an answer of the form $n \geq f(\varepsilon)$.

   Solution: Let p be the true proportion of the student body that is left-handed. If the dean wants a confidence interval of width ε for p, then he wants a margin of error $\frac{\varepsilon}{2}$. That is, if we let S_n be the number of students in a sample of n that are left-handed, he wants to guarantee that

$$P\left(\left|\frac{S_n}{n} - p\right| < \frac{\varepsilon}{2}\right) \geq 0.95.$$

   For this, it is sufficient that

$$0.95 \leq 2\Phi\left(2\left(\frac{\varepsilon}{2}\right)\sqrt{n}\right) - 1 = 2\Phi(\varepsilon\sqrt{n}) - 1.$$

   We solve for n to give the condition

$$n \geq \left(\frac{\Phi^{-1}(0.975)}{\varepsilon}\right)^2.$$

   Note: if you interpreted the question to mean that the margin of error should be ε, then we will give full credit. Under this assumption, you should end up with the inequality

$$n \geq \left(\frac{\Phi^{-1}(0.975)}{2\varepsilon}\right)^2.$$

   (b) (5 points) Suppose the dean's survey finds that the proportion of students that are left-handed is $p = 0.10$. Suppose that Professor Bob is teaching a 160 person Calculus class in a room with 20 left-handed desks. Use an appropriate approximation to estimate the probability that there will be more than 20 left-handed students in the class. (Assume that the hand preferences of different students are independent.)

   Solution: Let X_L be the number of left-handed students in Professor Bob's calculus class. If we assume that the hand preferences of different students are independent, then $X \sim \text{Bin}(n, p)$, where $n = 160$ and $p = \frac{1}{10}$. We want to estimate $P(X_L > 20)$. We plan to use the normal approximation for the binomial distribution; for this we must check that $np(1-p) \geq 10$. We see that

$$np(1-p) = 160\left(\frac{1}{10}\right)\left(\frac{9}{10}\right) = \frac{16 \times 9}{10} = \frac{12 \times 12}{10} > \frac{10 \times 12}{10} = 12 > 10.$$

   Thus, we can approximate $\frac{X-16}{12/\sqrt{10}}$ with a standard normal random variable. We have that

$$P(X > 20) = P\left(\frac{X-16}{12/\sqrt{10}} > \frac{20-16}{12/\sqrt{10}}\right) \approx 1 - \Phi\left(\frac{4}{12/\sqrt{10}}\right) = 1 - \Phi\left(\frac{\sqrt{10}}{3}\right).$$

(c) (5 points) Suppose that 0.1% (i.e. 1/1000) of the students in the university are ambidextrous. Suppose that the university admits a freshman class of $10,000$ students. Use an appropriate approximation to estimate the probability that exactly two of the new freshmen are ambidextrous.

Solution: Let X_A be the number of freshmen that are ambidextrous. Assuming that the hand preferences of different students are independent, we have that $X_A \sim \text{Binom}(m, q)$ with $m = 10000$ and $q = \frac{1}{1000}$. We want to compute $P(X_A = 2)$. We compute mq^2:

$$mq^2 = 10000\left(\frac{1}{1000}\right)^2 = \frac{10}{1000} = \frac{1}{100}.$$

Since $mq^2 \ll 1$, a Poisson approximation is appropriate. Specifically, if we let $Y \sim \text{Poisson}(mq)$, then we have

$$P(X_A = 2) \approx P(Y = 2) = e^{-10}\left(\frac{10^2}{2!}\right) = \frac{50}{e^{10}}.$$

(d) (3 points) We sample 40 students with replacement from the student body. Let X_1 be the number of times we choose a left-handed student, let X_2 be the number of times we choose an ambidextrous student, and let X_3 be the number of times we choose a right-handed student. Under the assumptions of parts (b)-(c), what distribution does the vector (X_1, X_2, X_3) have? (For full credit, you must specify all parameters.)

Solution: Letting p, q be as in parts (b)-(c), we have that

$$1 - p - q = \frac{899}{1000}.$$

Thus, the random vector (X_1, X_2, X_3) has the distrubution $\text{Multinom}\left(40, 3, \frac{1}{10}, \frac{1}{1000}, \frac{899}{1000}\right)$.

As an alternative to giving the name of the distribution, students may give the probability mass function: for any nonnegative integers k_1, k_2, k_3 with $k_1 + k_2 + k_3 = 40$, we have that

$$P(X_1 = k_1, X_2 = k_2, X_3 = k_3) = \binom{40}{k_1, k_2, k_3}\left(\frac{1}{10}\right)^{k_1}\left(\frac{1}{1000}\right)^{k_2}\left(\frac{899}{1000}\right)^{k_3}.$$

2. (10 points) Let X be a random variable with mean -4, and variance 16.
Find numbers $a, b > 0$ so that the random variable $Y = aX + b$ has mean 0 and variance 1.

Solution: The random variable $Z = X - (-4) = X + 4$ has mean 0. Note $Y = aZ$ has variance $\mathrm{Var}(Z) = a^2\mathrm{Var}(X) = a^2 16$. Choosing $a = 1/4$ ensures Y has variance one. Thus $b = 4a = 1$.

3. Let $X \sim \text{Unif}([-1,3])$. Compute the probability density functions of two the random variables:

(a) (10 points) $Y = X^3$

Solution: The function $g : x \mapsto x^3$ is strictly increasing, in particular on $[-1,3]$. Hence, the probability density functions f_Y of Y and f_X of X are related by the change of variables formula

$$f_Y(s) = f_X(g^{-1}(s))\left|\frac{\mathrm{d}}{\mathrm{d}s}g^{-1}(s)\right|$$

where s is in the range of Y given by $[g(-1), g(3)] = [-1, 27]$. Presently, $g^{-1}(s) = s^{1/3}$. Its derivative exists and equals $\frac{1}{3}s^{-2/3}$ as long as $s \neq 0$. If $s \in [-1, 27]$, then $f_X(g^{-1}(s)) = 1/4$. Consequently,

$$f_Y(s) = \frac{1}{12|s^{2/3}|}$$

if $s \in (-1, 27) \setminus \{0\}$, and $f_Y(s) = 0$ for any other s.

The lack of differentiability of the cumulative distribution function $F_Y(s) = F_X(s^{1/3})$ at $s = 0$ is irrelevant. The density functions f_Y can be assigned any value there. We chose put the f_Y to equal zero there for the ease of exposition.

(b) (10 points) $Y = |X|$

Solution: The range of Y is $[0,3]$. We compute its cumulative distribution function

$$F_Y(s) = P(Y \leq s) = P(X \in [-s, s]).$$

If $s \in [0,1]$, then $F_Y(s) = \int_{-s}^{s} \frac{1}{4} = s/2$. If $s \in (1,3]$, then $F_Y(s) = \int_{-1}^{s} \frac{1}{4} = s/4 + 1/4$. Differentiating yields that a probability density function f_Y of Y is

$$f_Y(s) = F_Y'(s) = \begin{cases} 1/2, & \text{if } 0 < s < 1, \\ 1/4, & \text{if } 1 \leq s \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

4. In this problem we consider the moment generating function of a discrete random variable X.

(a) (11 points) Suppose the moment generating function of X is

$$M_X(t) = \frac{1}{8} + \frac{3}{8}e^t + \frac{3}{8}e^{2t} + \frac{1}{8}e^{3t}.$$

Derive the probability mass function of X.

Solution: Reading off from the MGF we obtain $P(X = 0) = \frac{1}{8}, P(X = 1) = \frac{3}{8}, P(X = 2) = \frac{3}{8}, P(X = 3) = \frac{1}{8}$.

(b) (7 points) Use the derivatives of M_X to find the variance $\mathrm{Var}(X)$.

Solution: We must differntiatie. For the first moment we have:

$$\mathbb{E}X = M'(0) = \frac{3}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{8} = \frac{3}{2}$$

For the second moment we have:

$$\mathbb{E}X^2 = M''(0) = \frac{3}{8} + 2^2 \cdot \frac{3}{8} + 3^2 \cdot \frac{1}{8} = 3$$

And hence the variance is

$$\mathrm{var}(X) = \mathbb{E}X^2 - (\mathbb{E}X)^2 = \frac{3}{4}$$

(c) (1 point) Can you identify the distribution of X with parameters?

Solution: $\mathrm{Bin}(3, \frac{1}{2})$

5. Let (X, Y) be uniformly distributed on the triangle with vertices $(0,0)$, $(1,1)$, and $(1,2)$.

(a) (12 points) Find the marginals of the coordinates f_Y and f_X.

Solution: The area of the triangle is $1/2$, so the joint pdf is

$$f(x,y) = \begin{cases} 2 & 0 \le x \le 1, x \le y \le 2x \\ 0 & \text{otherwise} \end{cases}$$

The marginal pdfs are

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy = \begin{cases} \int_x^{2x} 2dy & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} 2x & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)dx = \begin{cases} \int_{x/2}^{x} 2dx & 0 \le y \le 1 \\ \int_{x/2}^{1} 2dx & 1 \le y \le 2 \\ 0 & y < 0 \text{ or } y > 2 \end{cases}$$

$$= \begin{cases} y & 0 \le y \le 1 \\ 2-y & 1 < y \le 2 \\ 0 & y < 0 \text{ or } y > 2 \end{cases}$$

(b) (3 points) Are the two coordinates independent or not?

Solution: They are not independent. Because $P(0 < X < 1/2) = 1/4$, $P(Y > 1) = 1/2$, but $P(0 < X < 1/2, Y > 1) = 0$.

6. Let $\lambda > 0$. Let $X \sim \text{Exp}(\lambda)$ and $Y \sim \text{Exp}(2\lambda)$ be independent exponentials.

(a) (9 points) Compute the probability $P(X > 2Y)$. Write the answer in terms of λ.

Solution: Because X and Y are independent, their joint pdf is

$$f(x,y) = f_X(x)f_Y(y) = \begin{cases} 2\lambda^2 e^{-\lambda x - 2\lambda y} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

So

$$P(X > 2Y) = \int_{x>2y} f(x,y)dxdy = \int_{x>2y>0} 2\lambda^2 e^{-\lambda x - 2\lambda y} dxdy$$

$$= \int_0^\infty \left(\int_{2y}^\infty \lambda e^{-\lambda x} dx \right) 2\lambda e^{-2\lambda y} dy = \int_0^\infty e^{-2\lambda y} 2\lambda e^{-2\lambda y} dy = \frac{2\lambda}{4\lambda} = 1/2$$

Alternatively, we see that $2Y \sim Exp(\lambda)$, so X and $2Y$ are i.i.d, hence $P(X > 2Y) = P(X < 2Y)$. And because they are jointly continuous, $P(X = 2Y) = 0$, hence $P(X > 2Y) = 1/2$.

(b) (9 points) Let $V = \min(X, Y)$. Derive a formula for the probability density function f_V of V.

Solution: Note that we have the following equality of events

$$\{V > v\} = \{X > v\} \cap \{Y > v\}.$$

Then by independence

$$\mathbb{P}(V > v) = \mathbb{P}(X > v) \cdot \mathbb{P}(Y > v) = e^{-\lambda v} e^{-2\lambda v} = e^{-3\lambda v}.$$

Therefore, the CDF and the PDF are given by

$$F_V(v) = 1 - e^{-3\lambda v} \qquad \Longrightarrow \qquad f_V(v) = 3e^{-3\lambda v}$$

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List of named distributions

$X \sim \text{Bernoulli}(p)$, PMF: $p_X(0) = 1 - p, \quad p_X(1) = p$

$X \sim \text{Binomial}(n, p)$, PMF: $p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad 0 \le k \le n$

$X \sim \text{Geometric}(p)$, PMF: $p_X(k) = p(1-p)^{k-1}, \quad k = 1, 2, \ldots$

$X \sim \text{Poisson}(\lambda)$, PMF: $p_X(k) = \frac{\lambda^k}{k!} e^{-\lambda}, \quad k = 0, 1, 2, \ldots$

$X \sim \text{Hypergeometric}(N, N_A, n)$, PMF: $p_X(k) = \frac{\binom{N_A}{k}\binom{N-N_A}{n-k}}{\binom{N}{n}}, \quad 0 \le k \le n$

$X \sim \text{Uniform}(a, b)$, PDF: $f_X(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b], \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Normal}(\mu, \sigma^2)$, PDF: $f_X(t) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}$

$X \sim \text{Exponential}(\lambda)$, PDF: $f_X(t) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$X \sim \text{Gamma}(r, \lambda)$, PDF: $f_X(t) = \begin{cases} \frac{1}{\Gamma(r)} \lambda^r x^{r-1} e^{-\lambda x} & \text{for } x > 0, \\ 0 & \text{otherwise} \end{cases}$

$(X_1, \ldots, X_k) \sim \text{Multinomial}(n, k, p_1, \ldots, p_k)$,
joint PMF: $p(a_1, \ldots, a_k) = \binom{n}{a_1, \ldots, a_k} p_1^{a_1} \cdots p_k^{a_k}$ for $a_1, \ldots, a_k \ge 0$ with $a_1 + \cdots + a_k = n$.

The following table shows approximate values of the cumulative distribution function of the standard normal distribution.

x	$\Phi(x)$
-2.0	0.022
-1.9	0.028
-1.8	0.035
-1.7	0.044
-1.6	0.054
-1.5	0.066
-1.4	0.080
-1.3	0.096
-1.2	0.115
-1.1	0.135
-1.0	0.158
-0.9	0.184
-0.8	0.211
-0.7	0.242
-0.6	0.274
-0.5	0.308
-0.4	0.344
-0.3	0.382
-0.2	0.420
-0.1	0.460
0.0	0.500
0.1	0.539
0.2	0.579
0.3	0.617
0.4	0.655
0.5	0.691
0.6	0.725
0.7	0.758
0.8	0.788
0.9	0.815
1.0	0.841
1.1	0.864
1.2	0.884
1.3	0.903
1.4	0.919
1.5	0.933
1.6	0.945
1.7	0.955
1.8	0.964
1.9	0.971
2.0	0.977